Frontend Development Portfolio

Rajesh Pattan

Hey! If you'd like to see these projects in action, I've posted screen recordings on my LinkedIn:

https://www.linkedin.com/in/rajesh-pattan-45453b3b4

A Little About This Portfolio

Hi there! This document is a walkthrough of the projects I built during my Frontend Web Development Internship. I created all of these from scratch, sticking closely to the 'Ponytail Philosophy': keeping things simple, clean, intentional, and highly functional.

To really push my understanding of how the web works, I purposely avoided heavy UI frameworks. Instead, I dove deep into core web technologies (HTML5, CSS3, and Vanilla JavaScript) to master DOM manipulation, performance optimization, and responsive design.

If you want to run these yourself:
1. Clone or download the repository to your local machine.
2. For most tasks, simply open the 'index.html' file in your browser.
3. For Task 4 (Spatial Portfolio), you'll need a quick local web server (like 'python -m http.server') to load the 3D assets.
4. That's it! No complex build tools or backend databases required.

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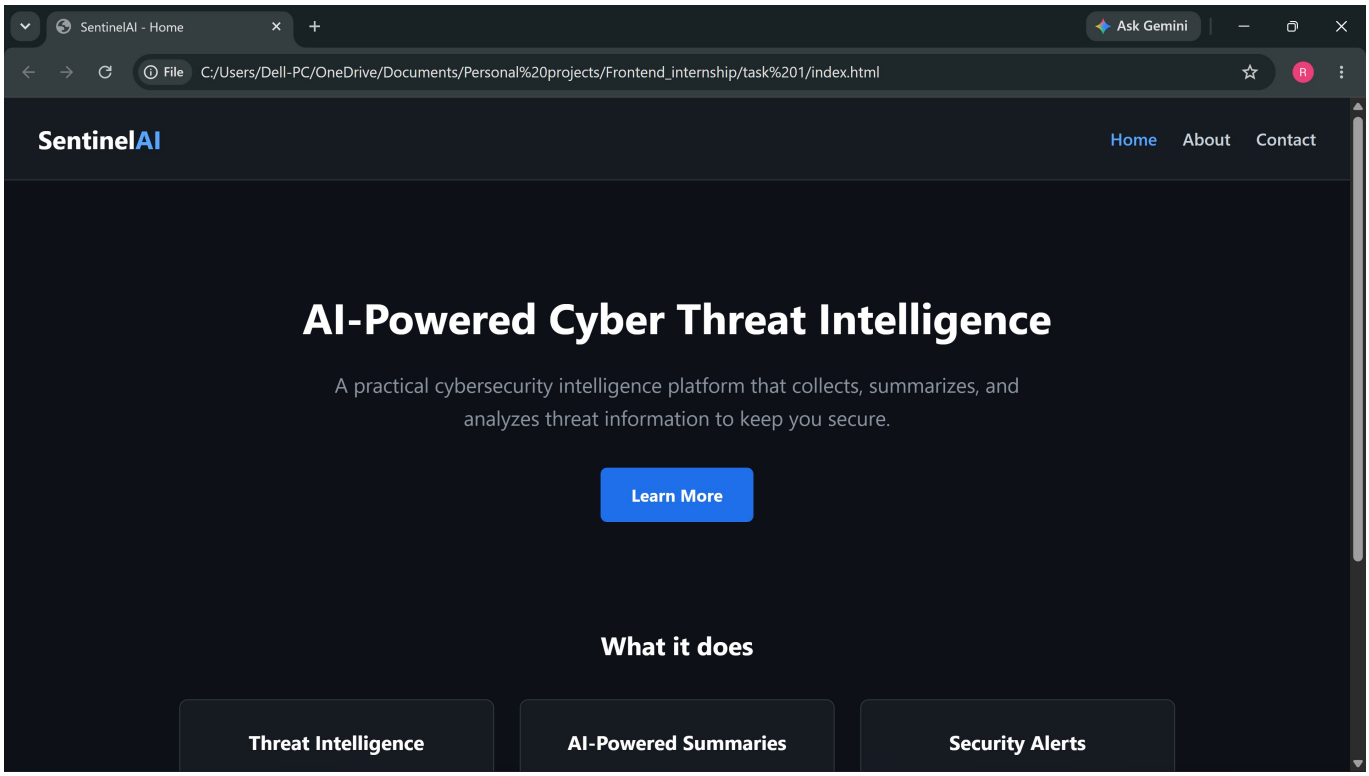
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Task 1: SentinelAI Build - Multi-Page Starter Website

What it is: A clean, minimal multi-page website for a fictional Cyber Threat Intelligence platform. I wanted to give it a strong cybersecurity aesthetic while focusing heavily on semantic structure.

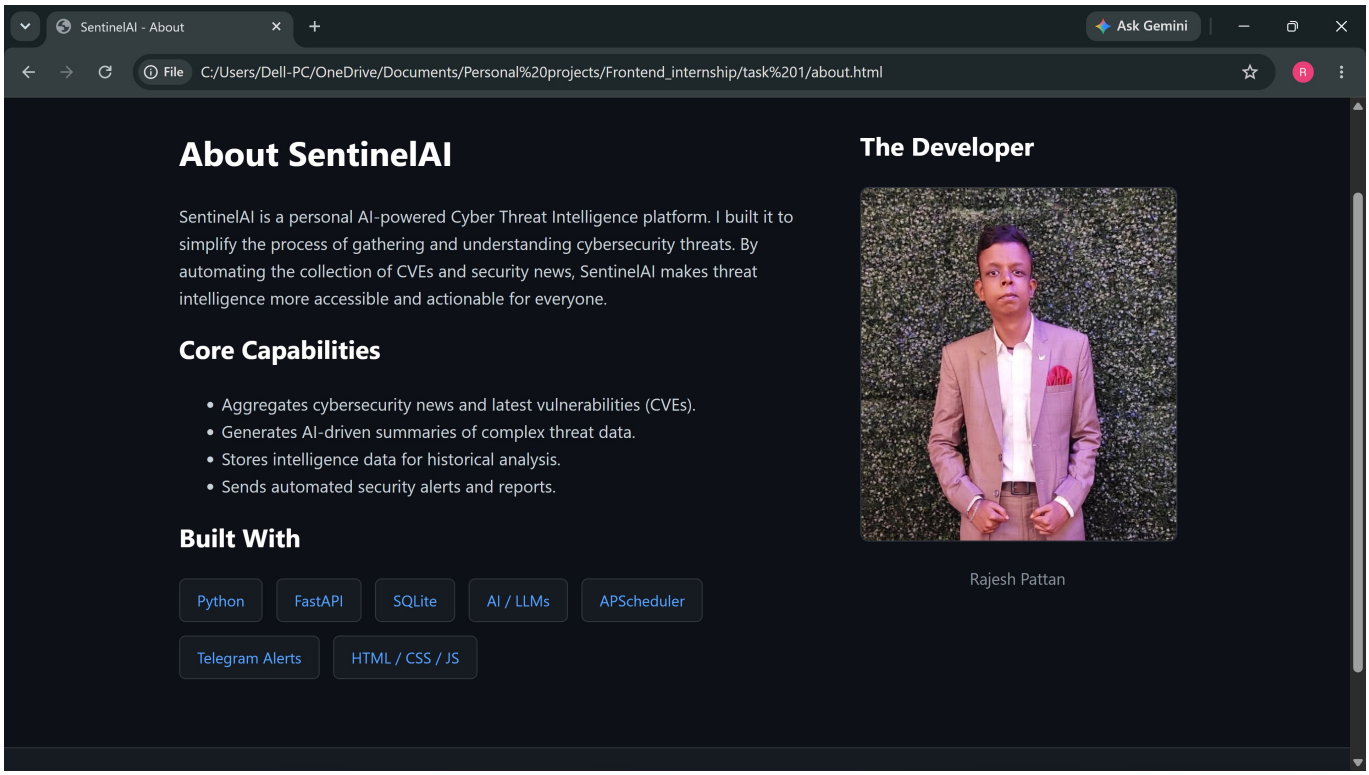
What I built:
- 3 fully linked HTML pages (Home, About, Contact).
- A responsive layout that looks great on desktop, tablet, and mobile.
- A contact form with basic client-side JavaScript validation.
- Strictly built with HTML5, CSS3, and Vanilla JavaScript.

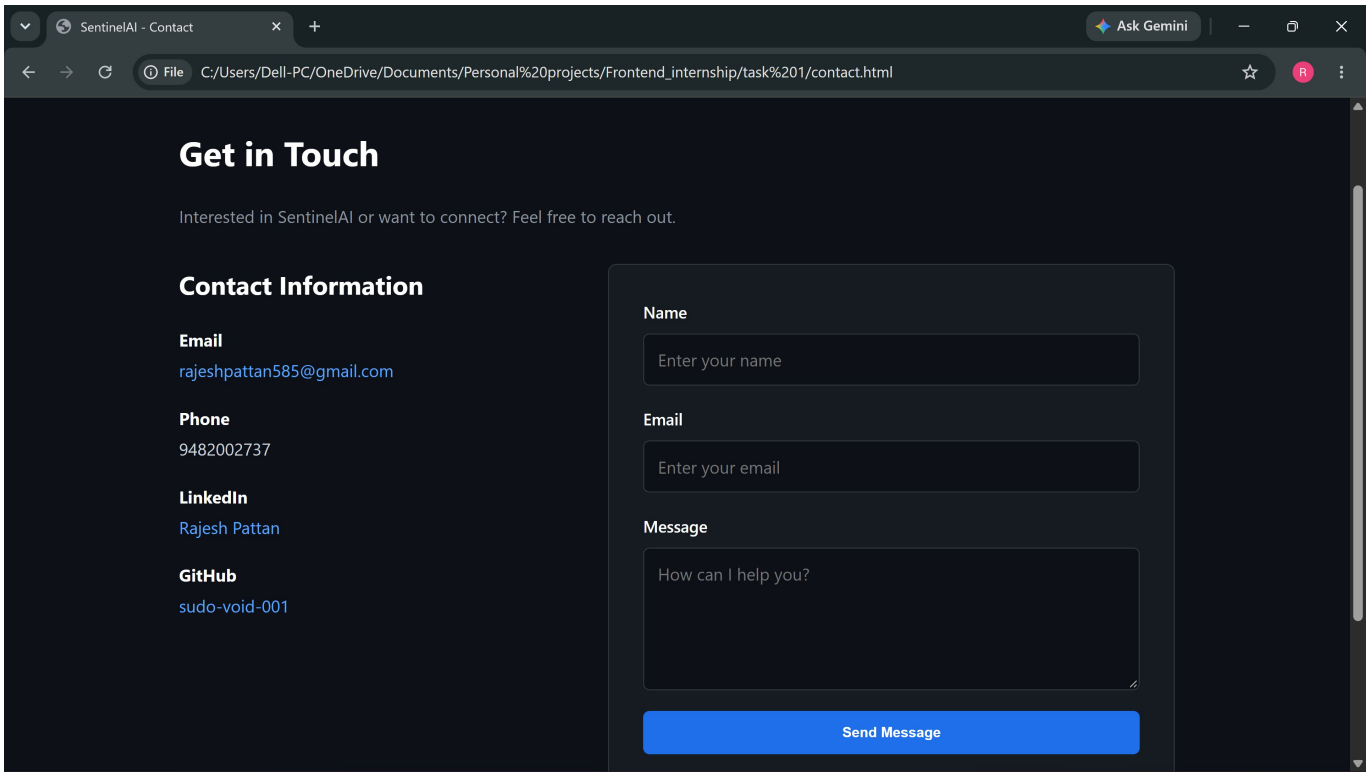
What I learned:
- How to properly structure semantic HTML for accessibility and SEO.
- Mastering modern layout modules like CSS Variables, Flexbox, and CSS Grid from the ground up.
- The fundamentals of routing across static HTML pages.



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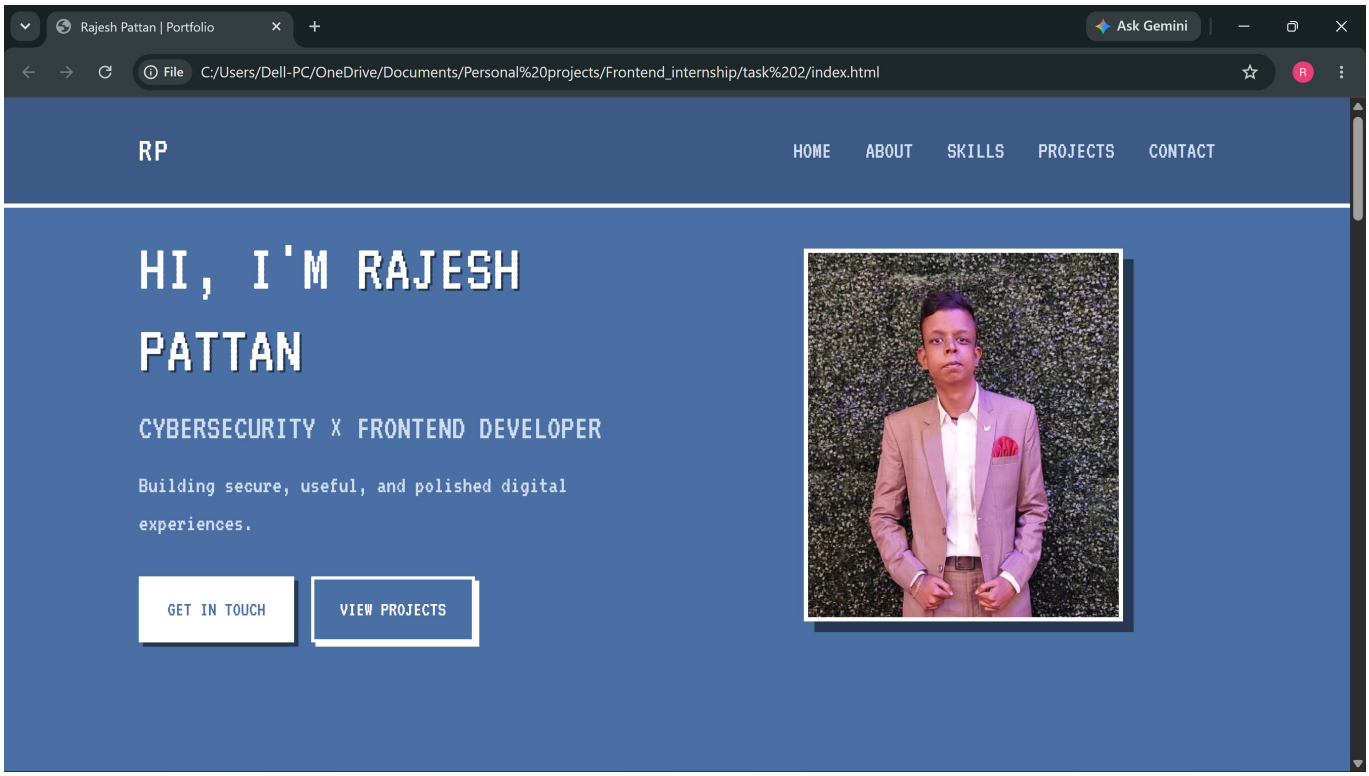
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Task 2: Personal Portfolio - Retro Edition

What it is: A single-page personal portfolio with a very distinct soft retro-terminal vibe. It uses VT323 pixel typography and a calm blue/white palette.

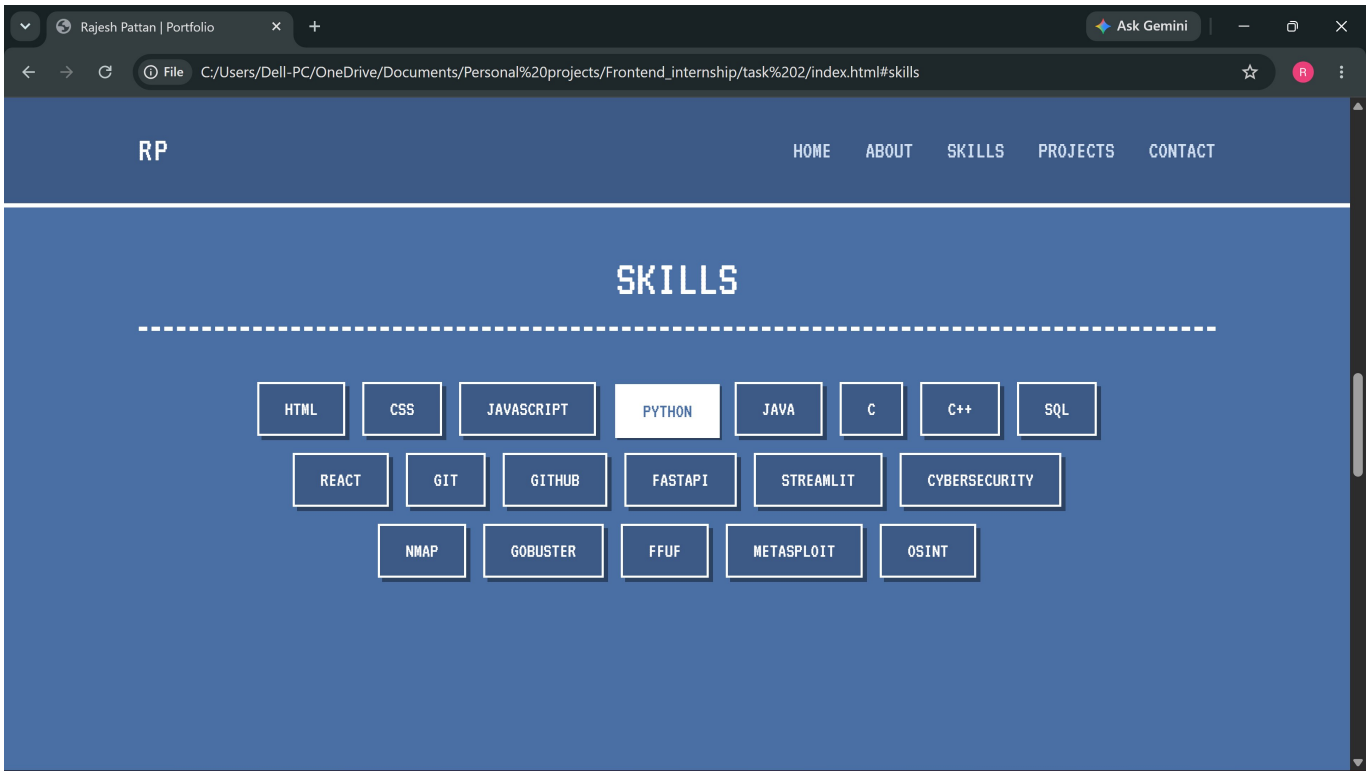
What I built:
- Scroll-triggered animations driven by the IntersectionObserver API.
- Interactive components with mechanical, tactile hover effects.
- A fully accessible mobile hamburger menu.
- Highly optimized CSS rendering that offloads animations to the GPU to prevent layout thrashing.

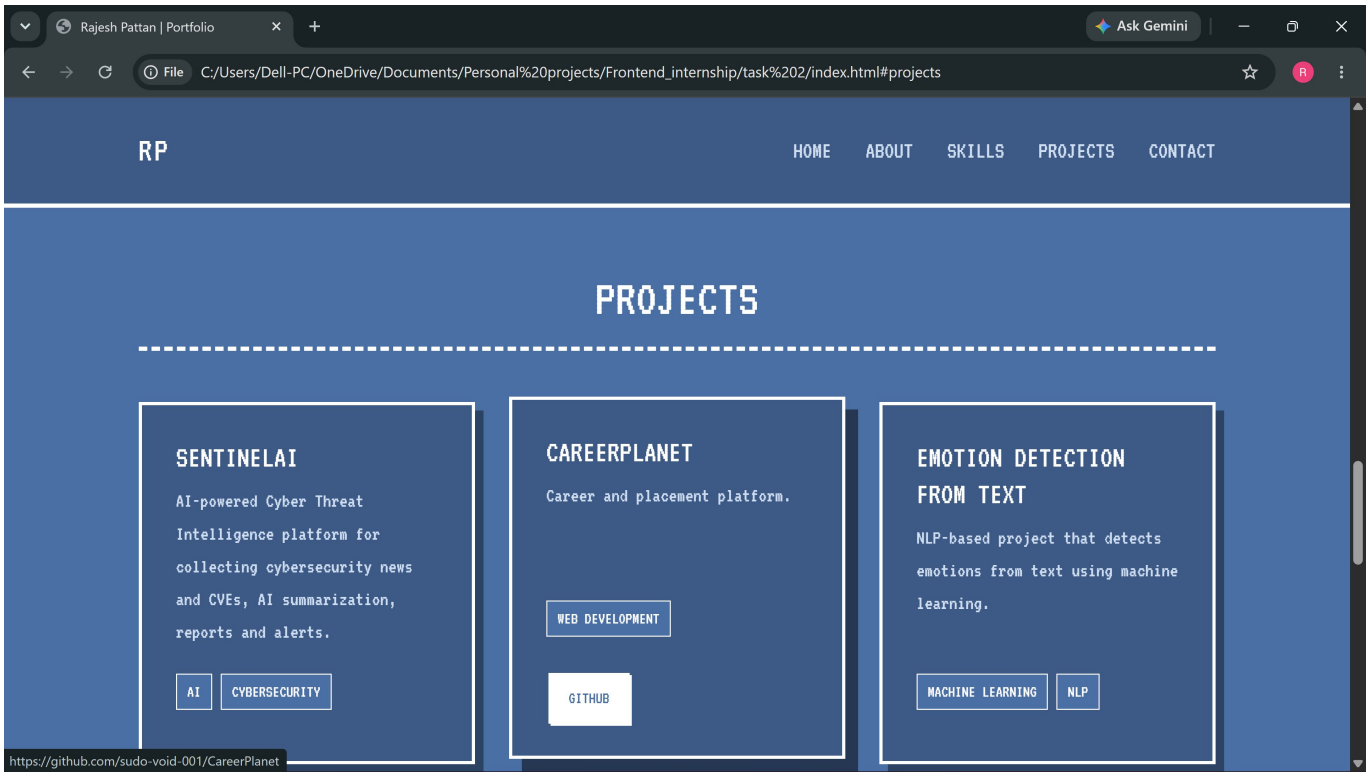
What I learned:
- Advanced ways to use Flexbox and Grid to build flexible project cards.
- Performance optimization techniques, like deferring scripts and leveraging 'will-change'.
- How to build and manage a custom CSS design system using CSS variables.



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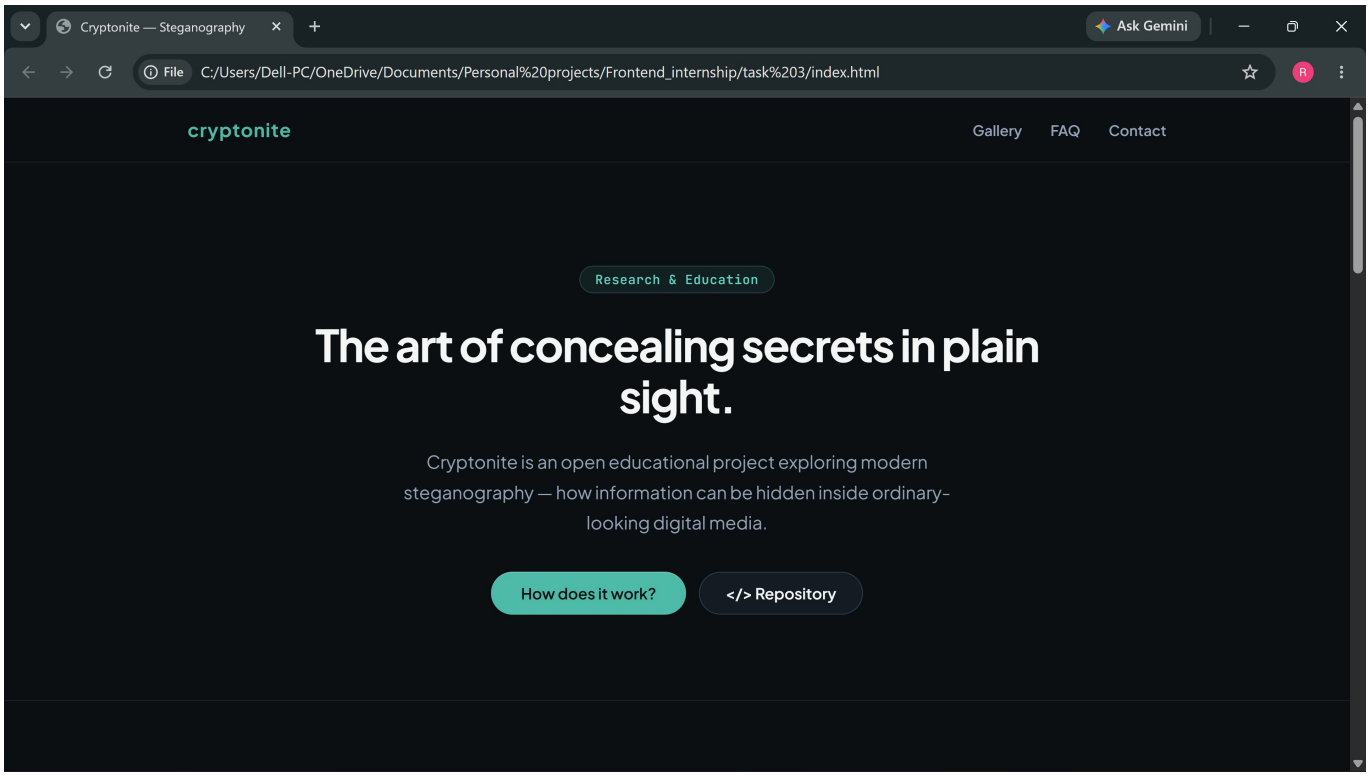
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Task 3: Cryptonite - JavaScript Interaction Web Page

What it is: A highly interactive Steganography-themed webpage designed to showcase complex DOM manipulations and event handling.

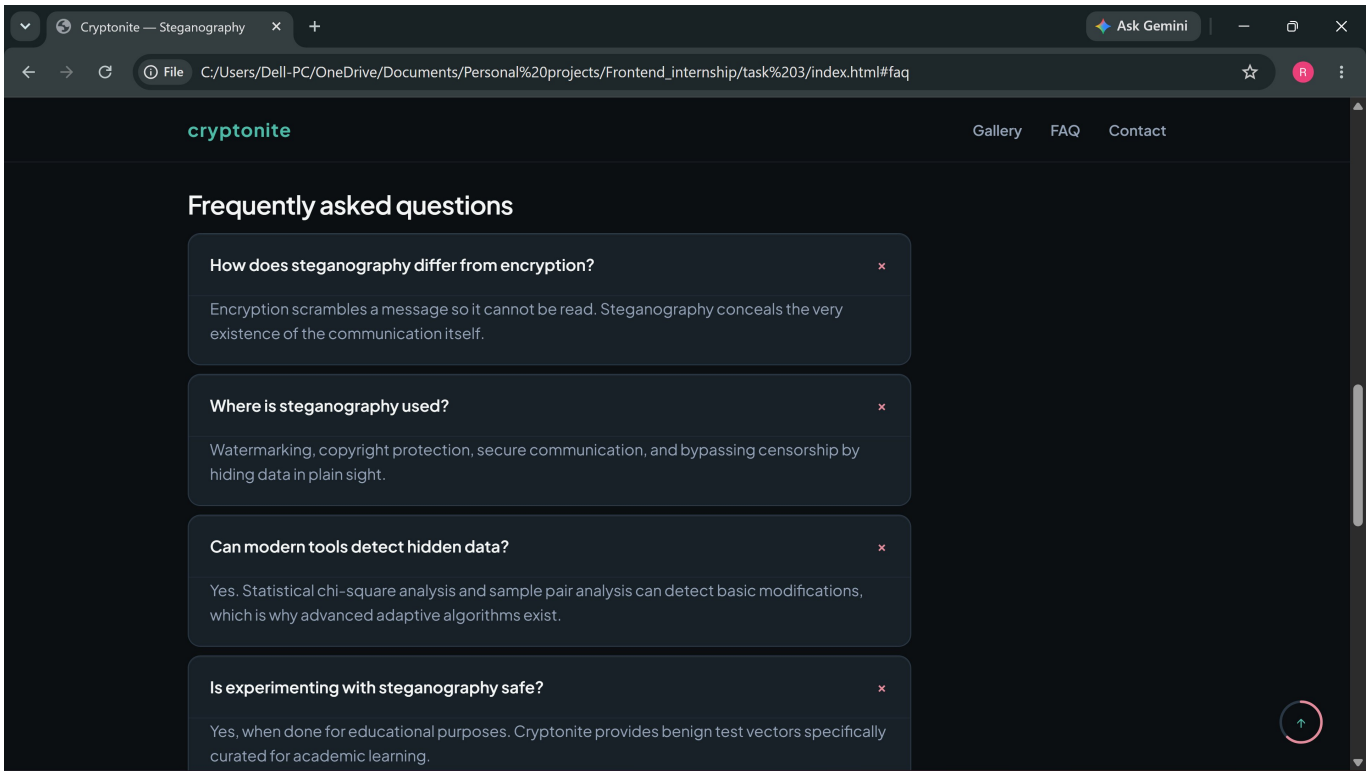
What I built:
- A side-scrolling image gallery that you can drag to navigate.
- An animated FAQ accordion and an informational modal popup.
- An SVG scroll progress ring that doubles as a back-to-top button.
- A randomized typewriter text effect for the hero title.
- Staggered scroll-reveal animations as you move down the page.

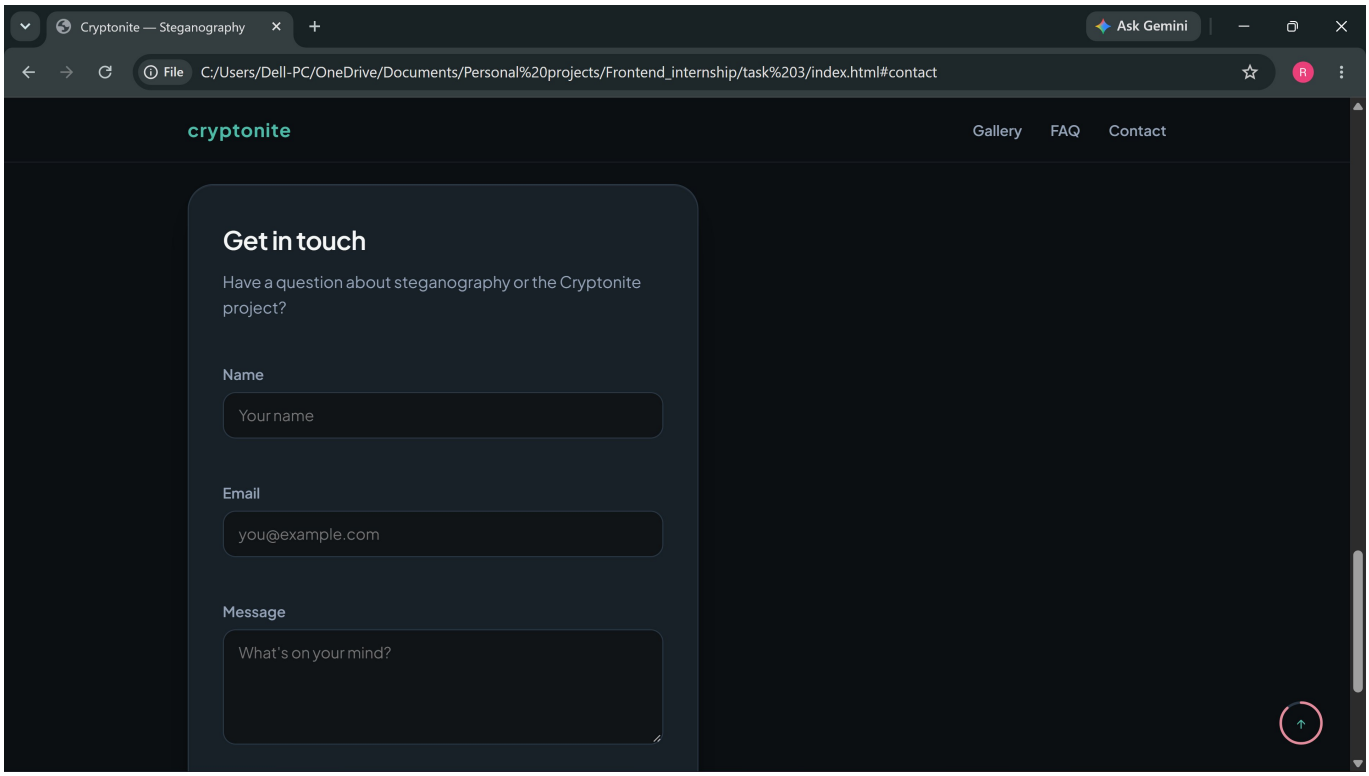
What I learned:
- Using extensive DOM selectors and attaching complex event listeners.
- Advanced manipulation of classes for modals, accordions, and validation states.
- Writing client-side regex for form input validation.



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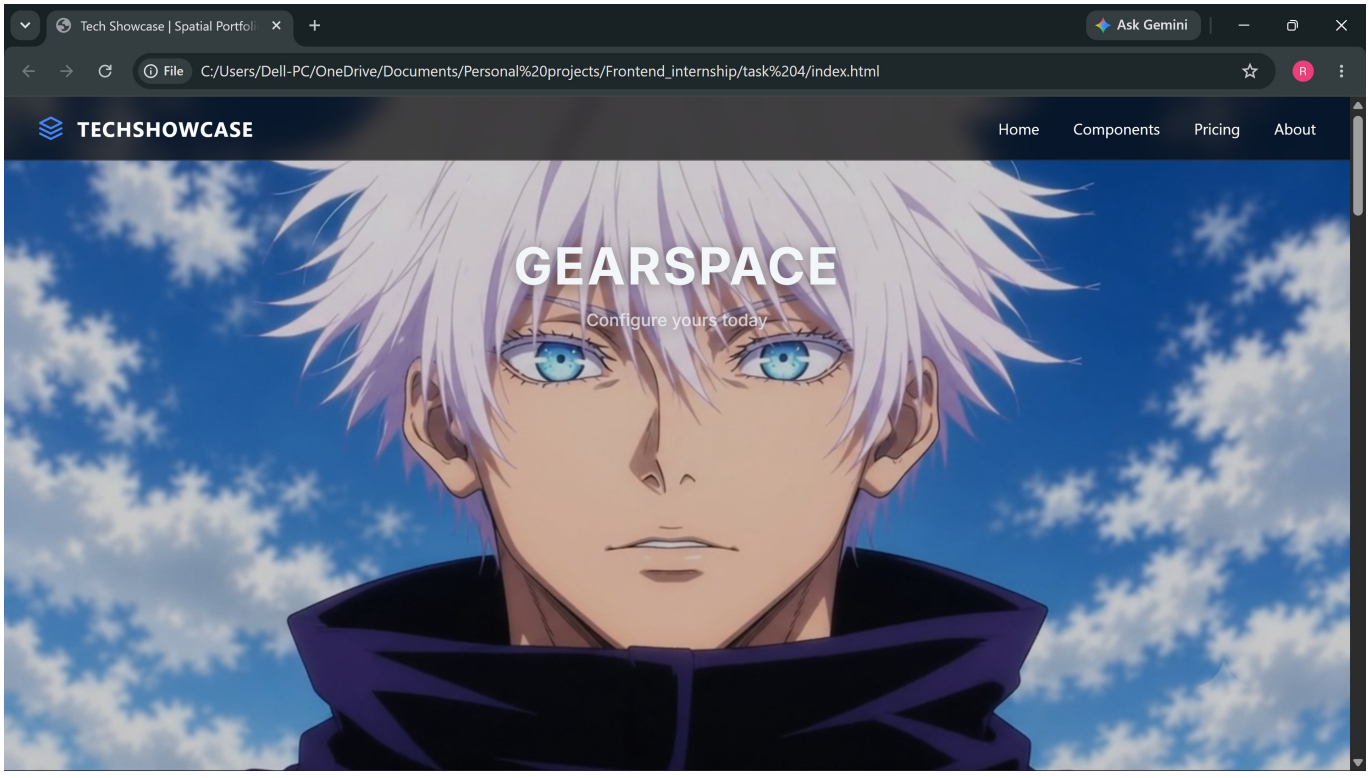
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Task 4: Tech Showcase - Spatial Portfolio

What it is: A cinematic, interactive technology showcase that blends standard web layouts with 3D hardware exploration to satisfy an intermediate service/business web page requirement.

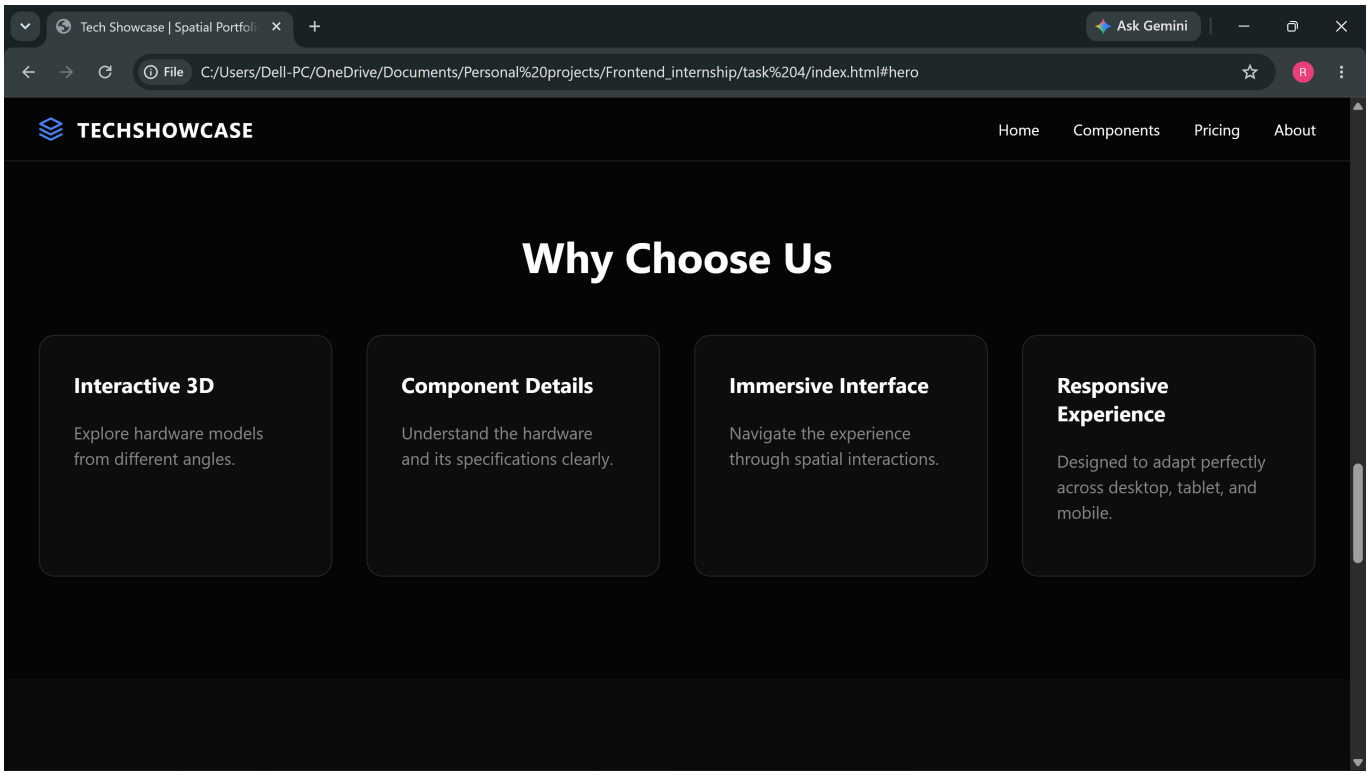
What I built:
- Interactive 3D laptop and hardware component previews rendered via Three.js.
- Cursor-following hero animations and edge-navigation controls triggered by cursor proximity.
- Translucent glassmorphism navigation and responsive pricing grids.

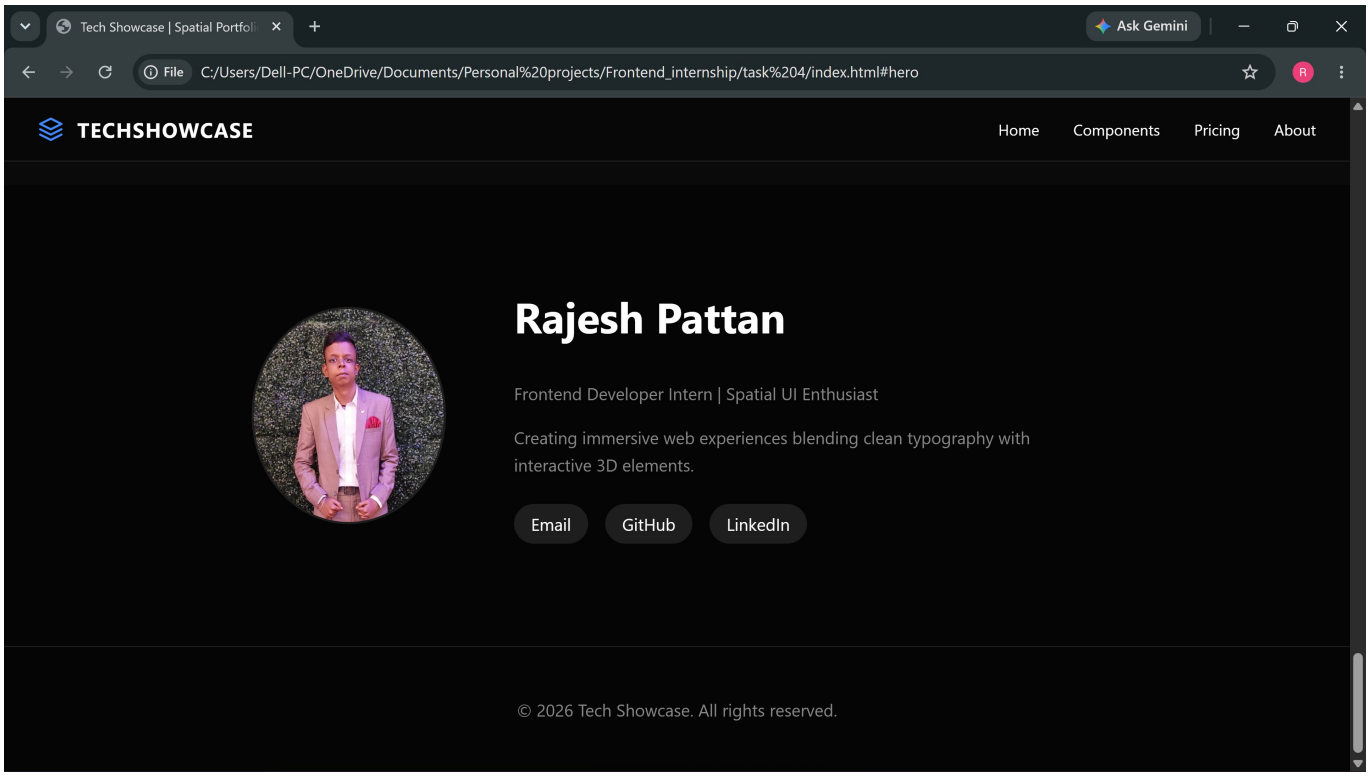
What I learned:
- Creating fluid structural adaptations across multiple breakpoints using CSS Grid and Flexbox.
- Using event listeners to compute cursor proximity and reveal UI elements dynamically.
- Safely integrating 3D GLB models into a traditional webpage layout without breaking scroll or overflow.



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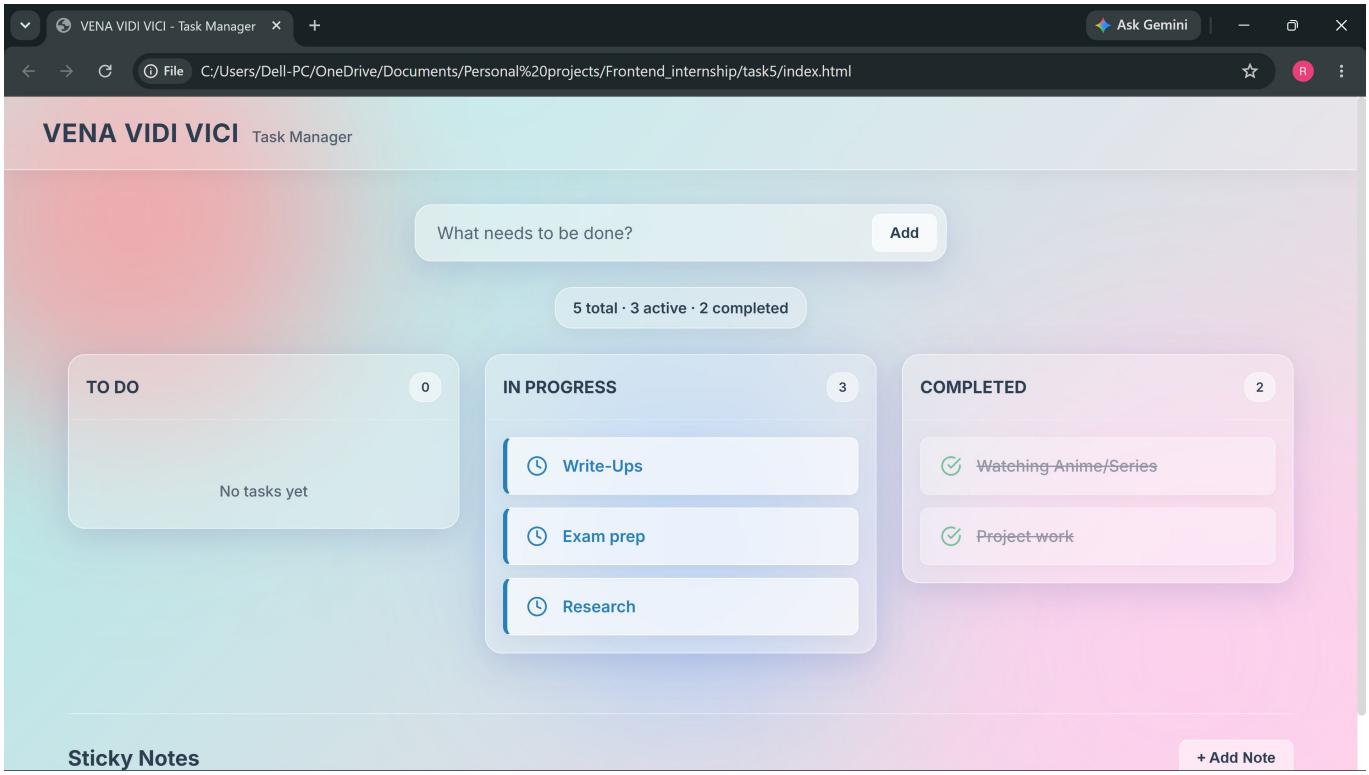
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Task 5: VENA VIDI VICI - Task Manager

What it is: A browser-based Task Manager built intuitively around Trello's visual column hierarchy.

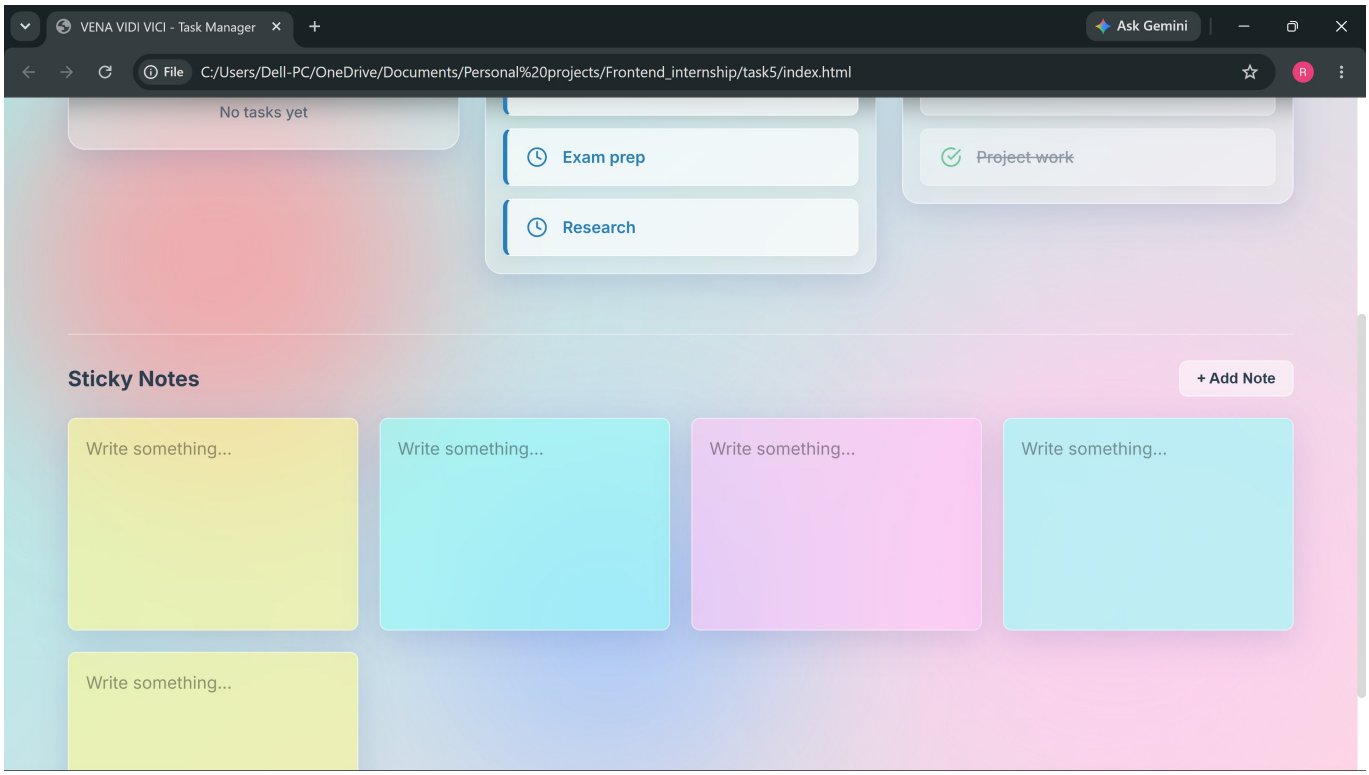
What I built:
- A dynamic board allowing users to add, cycle, and delete task cards across TODO, IN PROGRESS, and COMPLETED columns.
- Automatic data persistence so your tasks are still there when you refresh or close the browser.
- A fully responsive layout that adapts all the way down to mobile screens.

What I learned:
- How to generate and inject dynamic DOM elements using HTML templates and JavaScript.
- Managing client-side application state within a central JavaScript array.
- Utilizing browser localStorage to easily serialize and save data.



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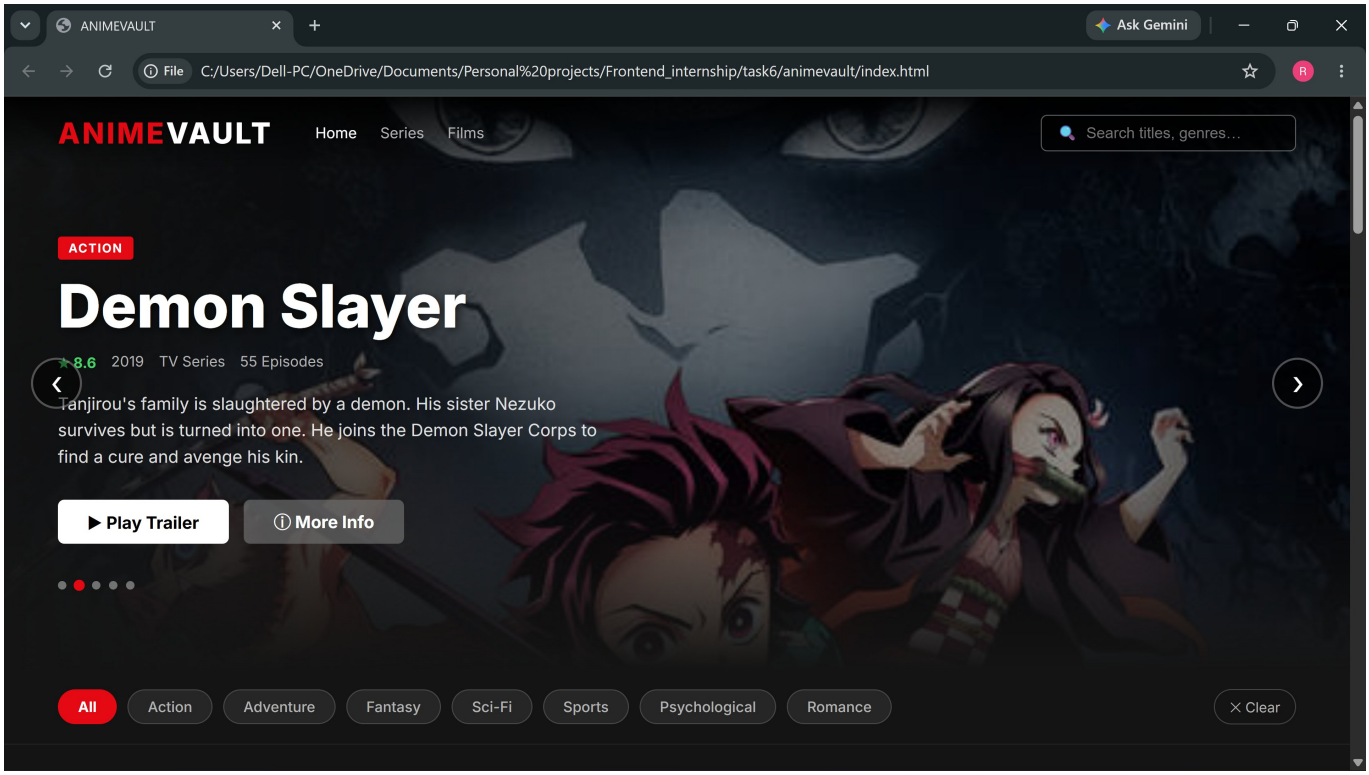
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Task 6: ANIMEVAULT - Interactive Anime Directory

What it is: A single-page directory application that dynamically renders a curated collection of anime, demonstrating robust filtering and search capabilities.

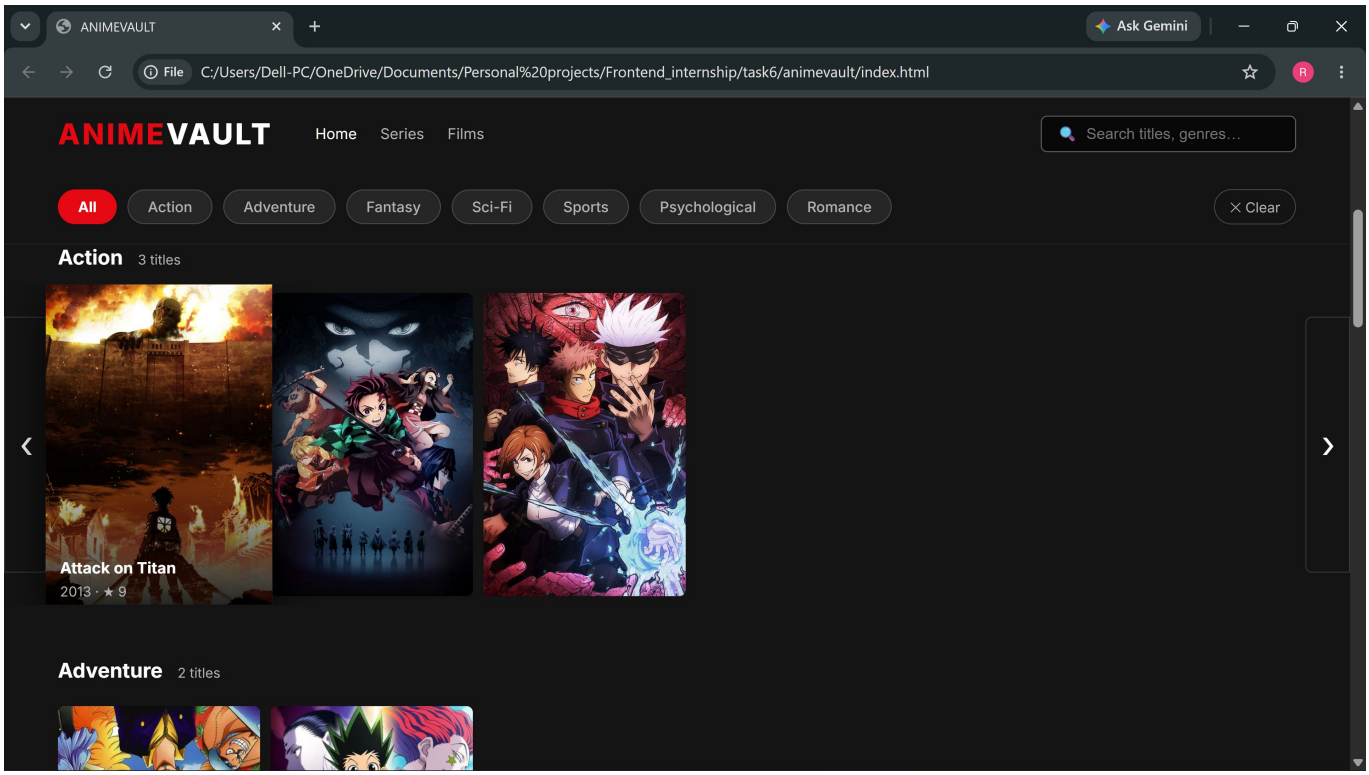
What I built:
- A grid of anime cards rendered entirely from a central JavaScript data object.
- Instant real-time search functionality and quick category filtering buttons.
- A rich detail modal presenting extensive metadata for any selected title.

What I learned:
- Treating JavaScript arrays/objects as a 'single source of truth' for rendering the UI.
- Building combined Search and Filter functions to create fluid, non-blocking user experiences.
- Managing complex UI states (like active filters and modals) entirely through Vanilla JavaScript.



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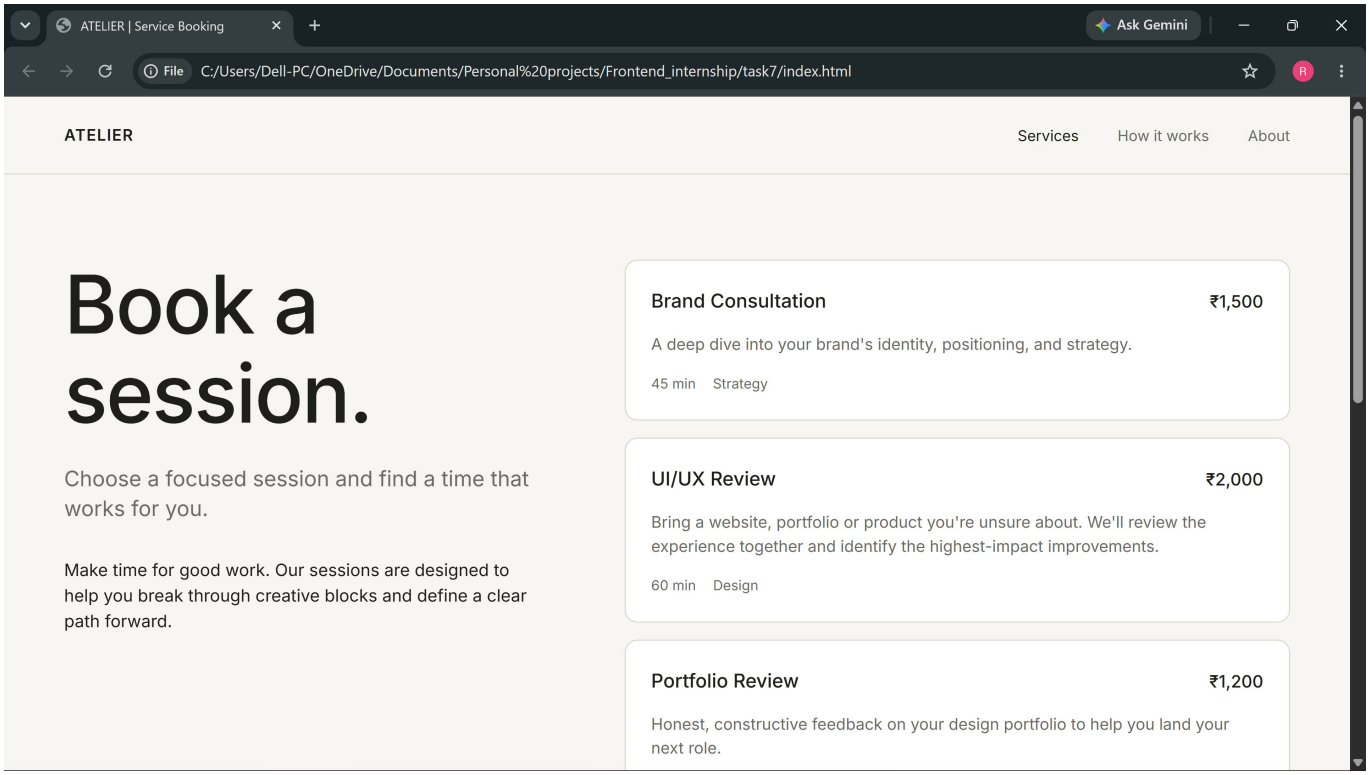
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Task 7: ATELIER - Service Booking Web Experience

What it is: A premium, minimalist multi-page booking application for a fictional creative studio, taking users gracefully through a step-by-step reservation process.

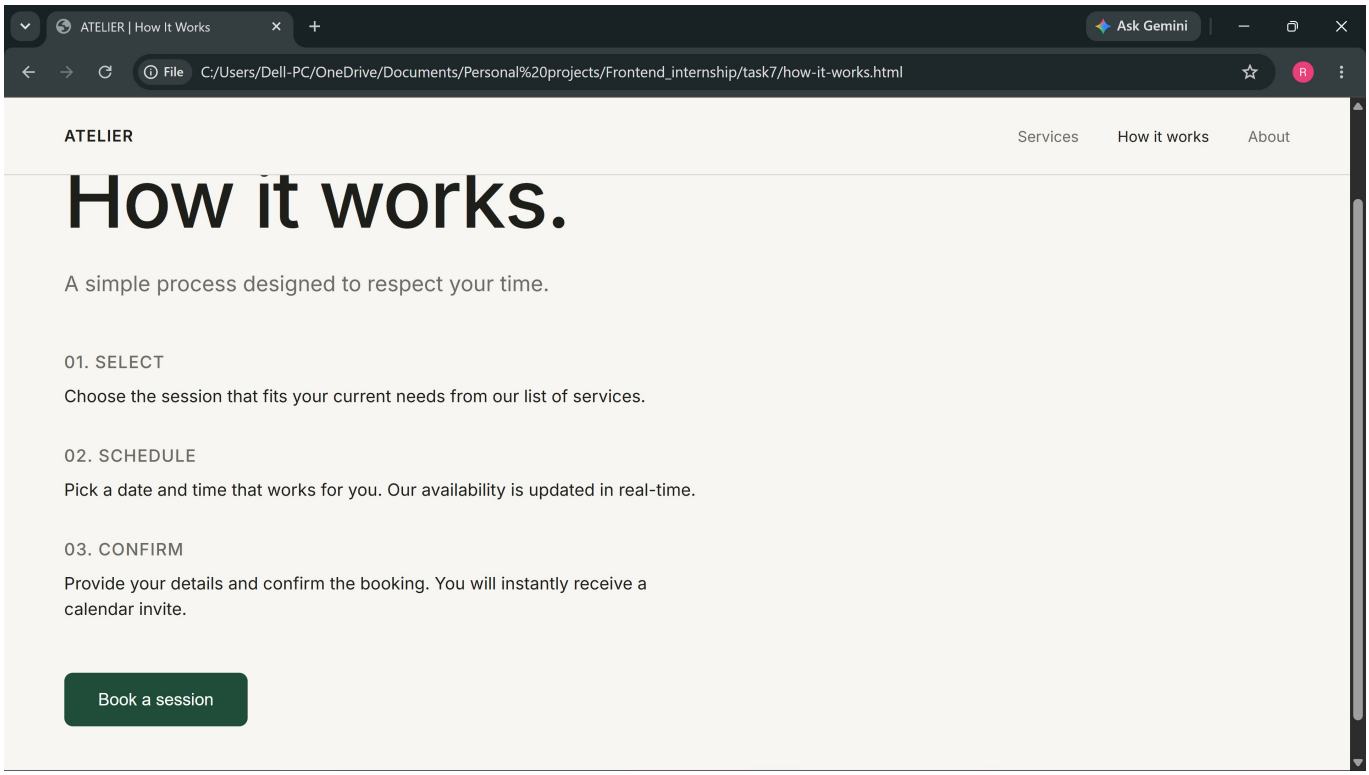
What I built:
- A step-by-step flow: Service Selection -> Date & Time -> Customer Details -> Review -> Confirmation.
- Persistent draft state handling so users don't lose booking progress if they accidentally refresh.
- Inline form validation providing immediate feedback to the user.

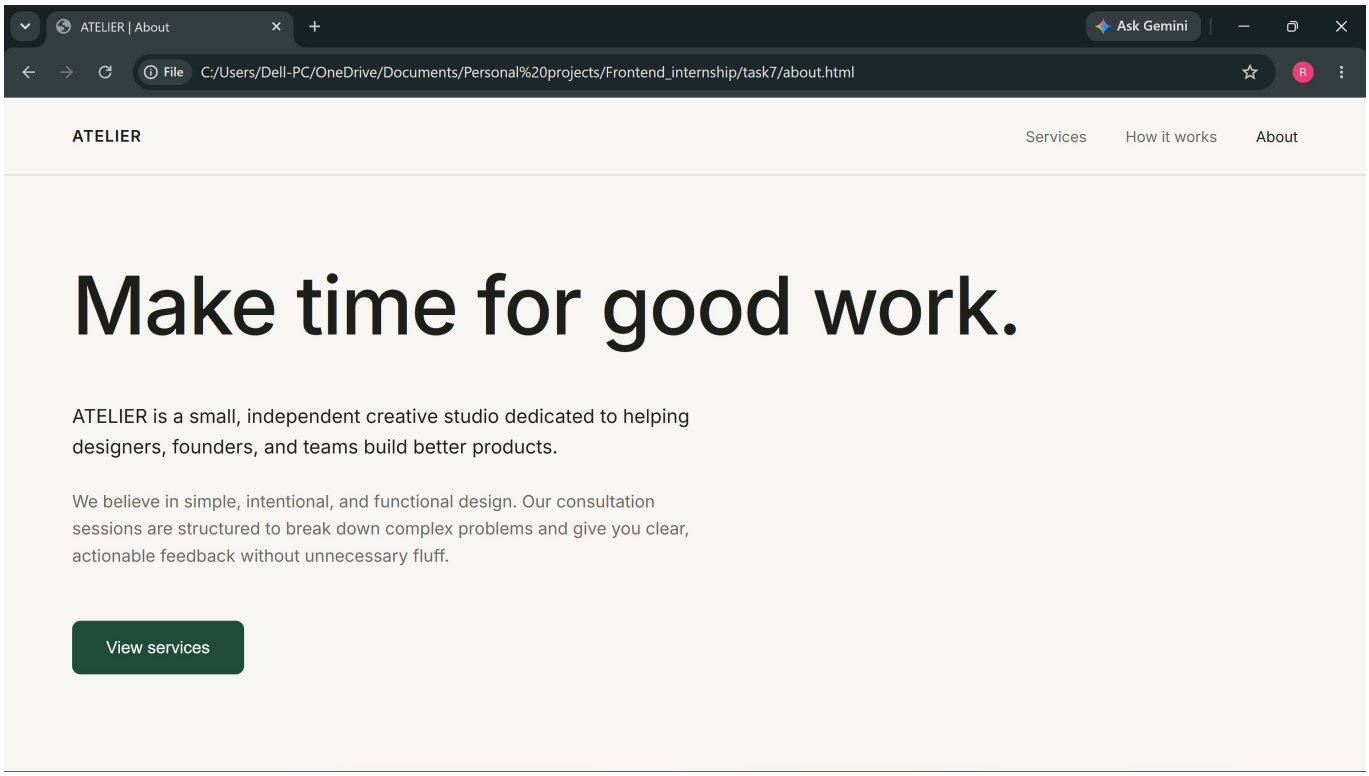
What I learned:
- Structuring a multi-page frontend architecture using progressive disclosure UI patterns.
- Propagating and keeping state in sync across different HTML pages utilizing localStorage.
- Building accessible and graceful interaction flows.



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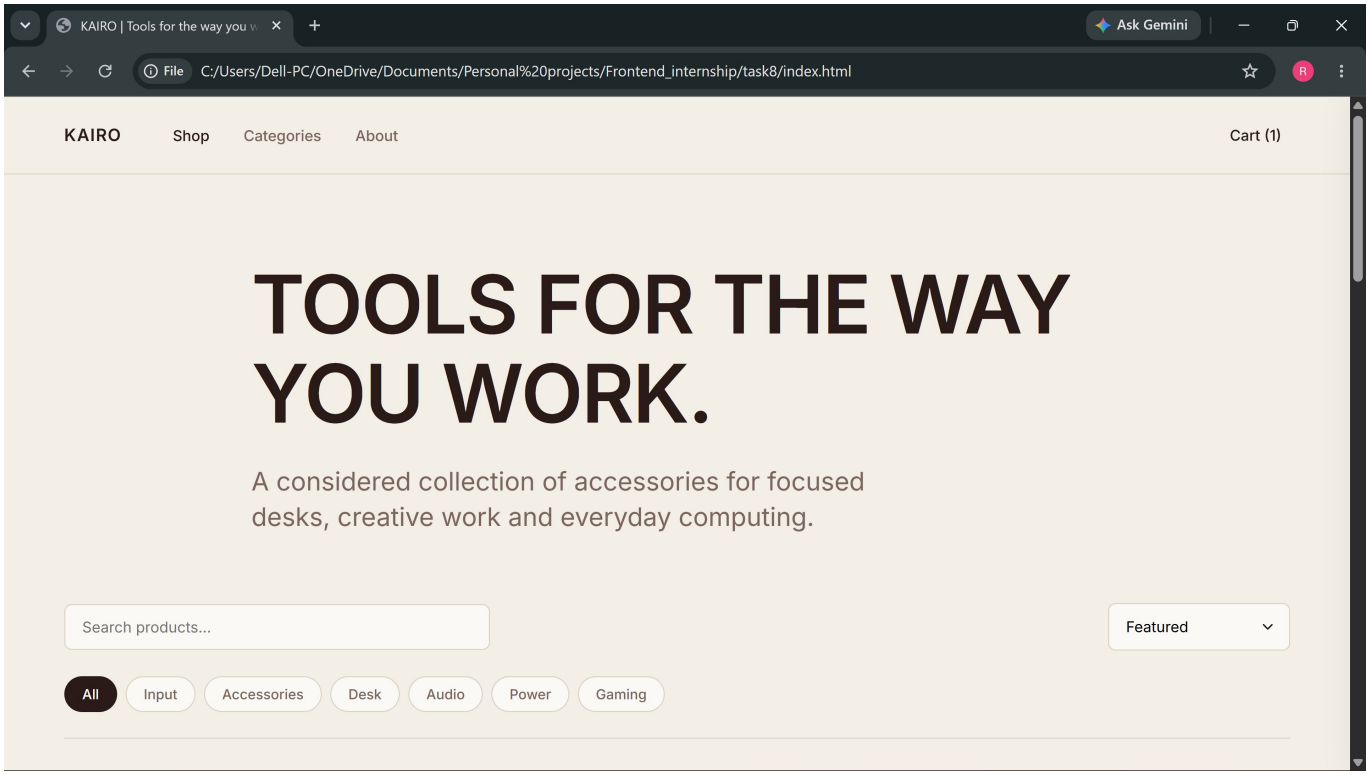
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Task 8: KAIRO - Product Catalog Web Experience

What it is: A robust e-commerce catalog showcasing modern accessories, emphasizing clean aesthetics and dynamic JavaScript rendering.

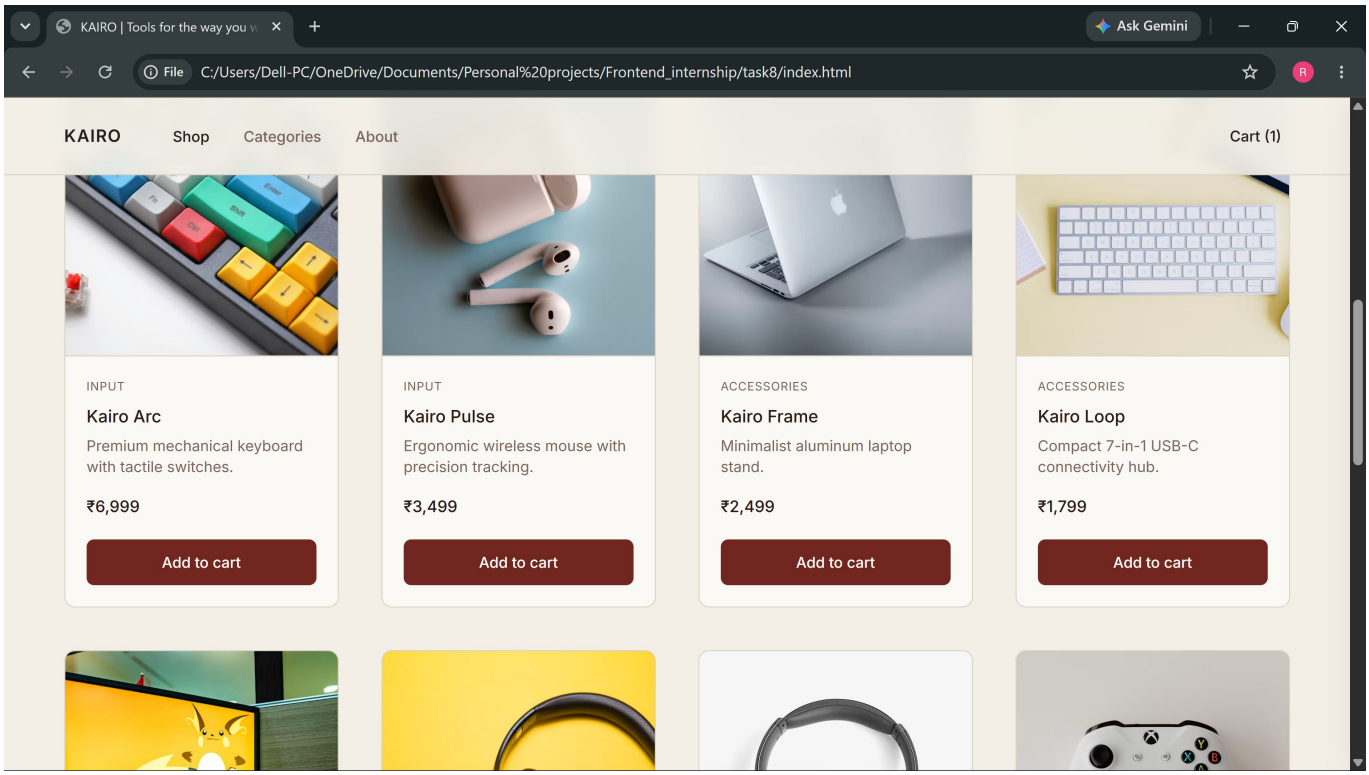
What I built:
- A dynamic product listing with individual detail pages routed purely via URL parameters (?id=...).
- A highly advanced pipeline combining Search, Category filtering, and Sorting algorithms seamlessly.
- A persistent, responsive slide-out Cart Drawer with item quantity management and subtotal calculation.

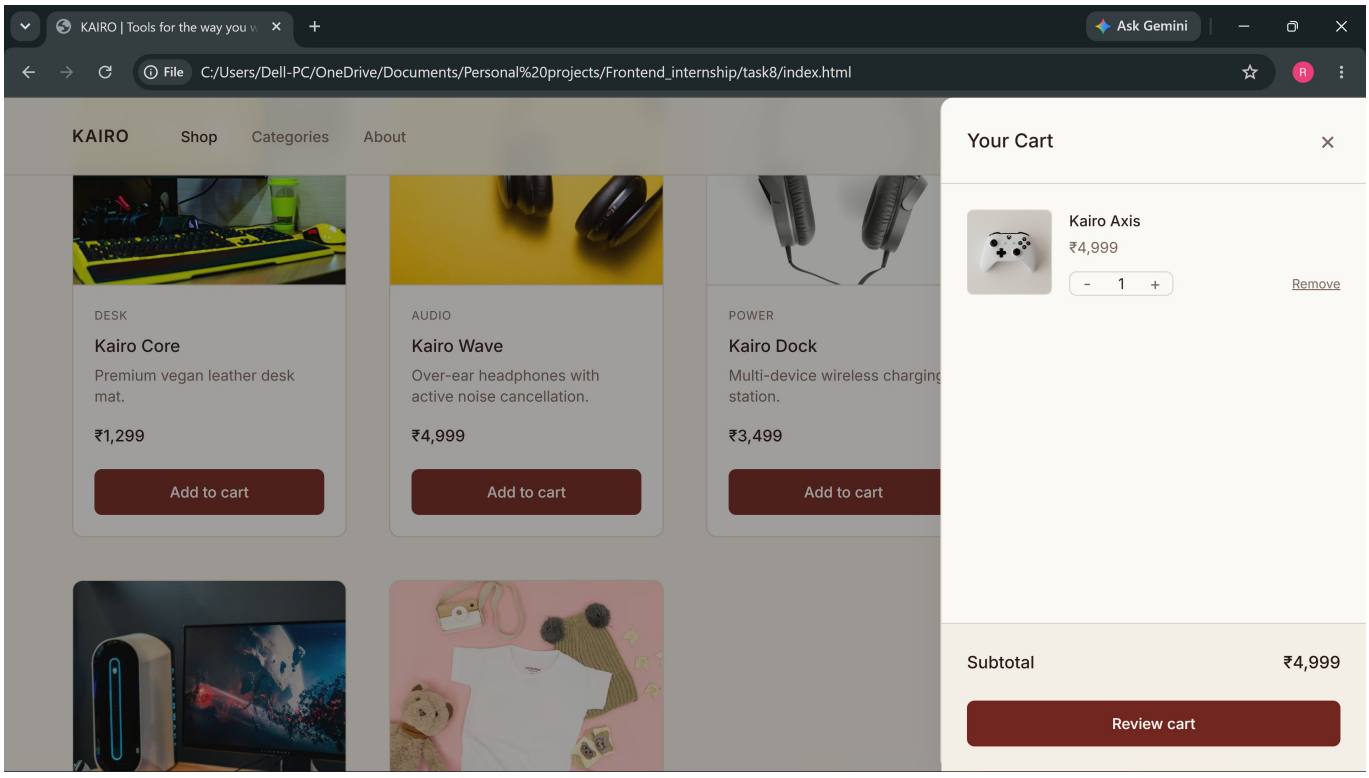
What I learned:
- Parsing URLSearchParams to pass contextual data between separate static HTML pages.
- Implementing complex state algorithms where search, filter, and sort operate harmoniously.
- Structuring a clean application architecture by separating logic across focused JavaScript modules.



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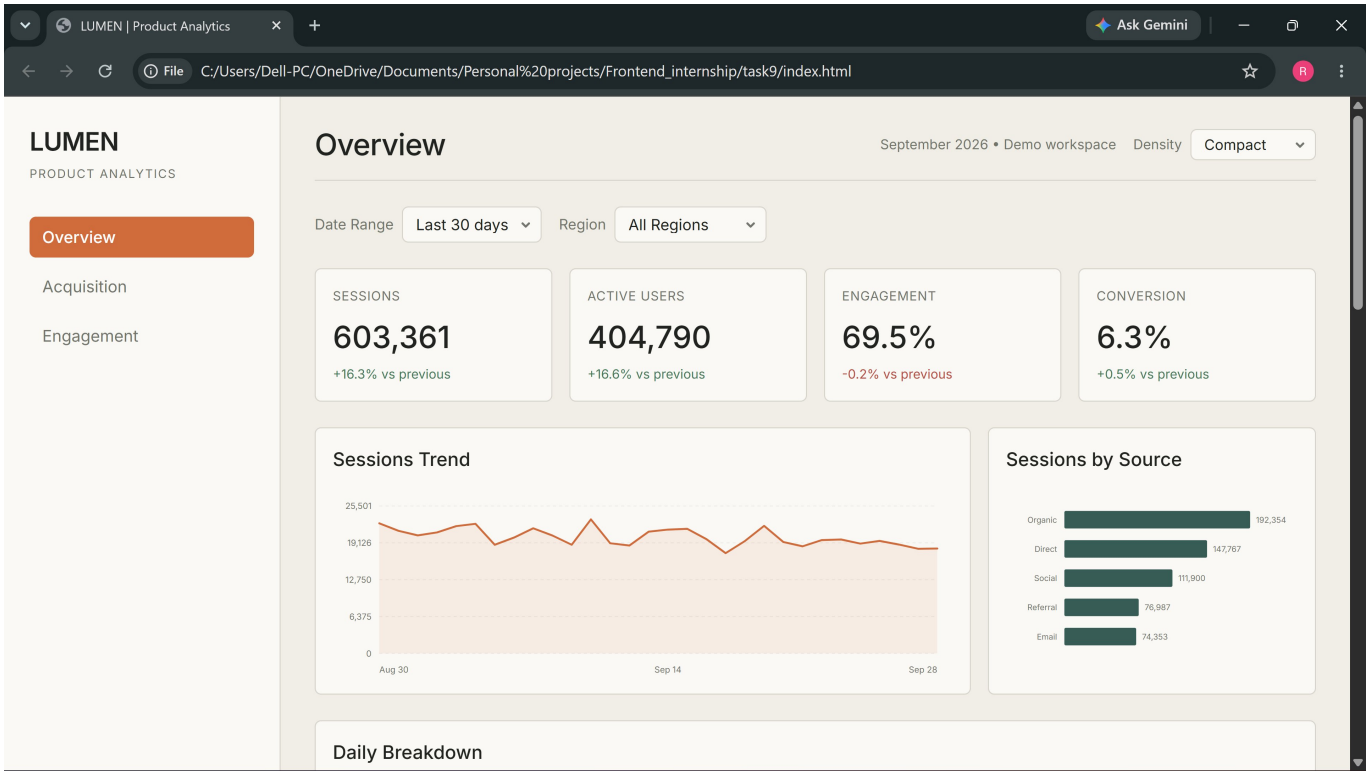
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Task 9: LUMEN - Interactive Analytics Dashboard

What it is: A sophisticated product analytics dashboard designed to visualize complex datasets clearly through custom DOM-manipulated charts.

What I built:
- A high-level KPI dashboard displaying primary metrics and percentage changes.
- Interactive regional and date-range filters that drive different data views (Overview, Acquisition, Engagement).
- Two dynamic vector charts (a Trend Line and a Bar Chart) drawn completely from scratch using SVG paths and polylines.
- Sortable data tables and a persistent UI density preference (Compact/Comfortable) saved via localStorage.

What I learned:
- Complex data-driven UI rendering involving client-side reductions and aggregations of raw datasets.
- Mathematical mapping and drawing of SVG elements directly via DOM manipulation, proving I can visualize data without relying on charting libraries.
- Designing scalable, accessible dashboard layouts that handle extensive UI state combinations.



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